



EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Lab Assignment, Spring 2026 Semester

Course: CSE207 (Data Structures)

Instructor: Puja Chakraborty, Senior Lecturer, CSE Department, EWU.

Full Marks: 10

Topic: Linked List

- 1) Write a pseudocode to **insert an element at the first position** of the linked list.
- 2) Write a pseudocode to **insert an element at kth position** of the linked list.
- 3) Write a pseudocode to **insert an element at the last position** of a linked list.
- 4) Write a pseudocode **to insert an element to a sorted (in ascending order)** linked list so that it still remains sorted.
- 5) Write a pseudocode to **delete duplicate elements** from a sorted (in ascending order) linked list.
- 6) Write pseudocode to check whether **two singly linked lists are identical**.
- 7) Write pseudocode to **delete the entire linked list** and free memory. (Destructor)
- 8) Write pseudocode to **merge two singly linked lists**.
- 9) Write pseudocode to **check whether a singly linked list is sorted or not**.
- 10) Write a pseudocode to **print a singly linked list**.

Follow the given instructions to build a linked list in C++

- 1) Design a **template class** `Node<T>` that contains the following data members:
 - `data` of type `T`
 - `next`, a pointer to another `Node<T>`
- 2) Design a **template class** `LinkedList<T>` to represent a singly linked list. The class should maintain a pointer named `head` that points to the first node of the list.
- 3) Implement the following member functions in the `LinkedList<T>` class:
 - Insert an element at the **k-th position** in the linked list
 - Delete a specified element from the linked list
 - Display (print) all elements of the linked list
- 4) Implement a **destructor** for the `LinkedList<T>` class to properly deallocate all dynamically allocated memory.
- 5) Use **constructors wherever necessary** to initialize class data members.
- 6) In the `main()` function, create an object of the `LinkedList<T>` class and demonstrate the use of the implemented operations.

Submission Instructions:

1. You will need to submit a PDF file that should contain the pseudocode (either in typed format or hand-written and then you can paste the photo).
2. Add the screenshots of your code segment by segment as the instruction given above.